

AIM:- Write a program to implement Tic-Tac-Toe using python

CODE:- import random

```
def drawBoard(board):
    #Prints the current board (ignore index 0)
    print(' | |')
    print(' ' + board[7] + ' | ' + board[8] + ' | ' + board[9])
    print(' | |')
    print('-----')
    print(' | |')
    print(' ' + board[4] + ' | ' + board[5] + ' | ' + board[6])
    print(' | |')
    print('-----')
    print(' | |')
    print(' ' + board[1] + ' | ' + board[2] + ' | ' + board[3])
```

```
def inputPlayerLetter():
    letter = ''
    while not (letter == 'X' or letter == 'O'):
        print('Do you want to be X or O?')
        letter = input().upper()
    if letter == 'X':
        return ['X', 'O']
    else:
        return ['O', 'X']
```

```
def whoGoesFirst():
    if random.randint(0, 1) == 0:
        return 'computer'
    else:
        return 'player'
```

```
def playAgain():
    print("Do you want to play again? (yes or no)")
    return input().lower().startswith('y')
```

```
def makeMove(board, letter, move):
    board[move] = letter
```

```
def isWinner(bo, le):
    # Given board and letter, return True
    # if player has won
    return (
        (bo[7] == le and bo[8] == le
         and bo[9] == le) or
        (bo[4] == le and bo[5] == le
         and bo[6] == le) or
        (bo[1] == le and bo[2] == le
         and bo[3] == le) or
        (bo[7] == le and bo[4] == le
         and bo[1] == le) or
        (bo[8] == le and bo[5] == le
         and bo[2] == le) or
        (bo[9] == le and bo[6] == le
         and bo[3] == le) or
        (bo[7] == le and bo[5] == le
         and bo[3] == le) or
```


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```
(b0[9]==le and b0[5]==le and
b0[1]==le)
```

```
def getBoardCopy(board):
    return board[:]
```

```
def isSpaceFree(board, move):
    return board[move]!=''
```

```
def getPlayerMove(board):
    move=' '
    while move not in '1 2 3 4 5 6 7 8 9'
        .split() or not isSpaceFree(board,
        int(move)):
        print('What is your next move?(1-9)')
        move=input()
    return int(move)
```

```
def chooseRandomMoveFromList(board, movesList):
    possibleMoves=[m for m in movesList if
        isSpaceFree(board, m)]
    if possibleMoves:
        return random.choice(possibleMoves)
    else:
        return None
```

```
def getComputerMove(board, computerLetter):
    if computerLetter=='X':
        playerLetter='O'
    else:
        playerLetter='X'
```

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```
for i in range(1, 10):
    copy = getBoard(copy(board))
    if isSpaceFree(copy, i):
        makeMove(copy, computerLetter, i)
        if isWinner(copy, computerLetter):
            return i
```

```
for i in range(1, 10):
    copy = getBoard(copy(board))
    if isSpaceFree(copy, i):
        makeMove(copy, playerLetter, i)
        if isWinner(copy, playerLetter):
            return i
```

```
for i in range(1, 10):
    copy = getBoard(copy(board))
    if isSpaceFree(copy, i):
        makeMove(copy, playerLetter, i)
        if isWinner(copy, playerLetter):
            return i
```

```
move = chooseRandomMoveFromList(board, [1, 3, 7, 9])
if move:
    return move
```

```
if isSpaceFree(board, 5):
    return 5
```

```
return chooseRandomMoveFromList(board, [2, 4, 6, 8])
```

Sign of Faculty with Date

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```
def isBoardFull(board):
    for i in range(1, 10):
        if isSpaceFree(board, i):
            return False
    return True
```

```
print('Welcome to Tic Tac Toe!')
```

```
while True:
    theBoard = [' '] * 10
    playerLetter, computerLetter = inputPlayerLetter()
    turn = whoGoesFirst()
    print('The ' + turn + ' will go first.')
    gameIsPlaying = True
```

```
while gameIsPlaying:
    if turn == 'player':
        drawBoard(theBoard)
        move = getPlayerMove(theBoard)
        makeMove(theBoard, playerLetter, move)

        if isWinner(theBoard, playerLetter):
            drawBoard(theBoard)
            print('Hooray! You have won the game!')
            gameIsPlaying = False
        else:
            if isBoardFull(theBoard):
                drawBoard(theBoard)
                print('The game is a tie!')
                break
            else:
                turn = 'computer'
```

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else: ~~else:~~

move = getComputerMove(theBoard, computerLetter)
makeMove(theBoard, computerLetter, move)

if isWinner(theBoard, computerLetter):

drawBoard(theBoard)

print('The computer has beaten you!
You lose.')

gameIsPlaying = False

else:

if isBoardFull(theBoard):

drawBoard(theBoard)

print('The game is a tie!')

break

else:

turn = 'player'

if not playAgain():
break

OUTPUT:

Welcome to Tic Tac Toe

Do you want to be X or O?

X

The player will go first

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What is your next move? (1-9)

9

0		X

What is your next move? (1-9)

5

0		X
	X	
0		

What is your next move? (1-9)

4

0		X
X	X	0
0		

What is your next move? (1-9)

2

0	0	X
X	X	0
0	X	

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What is your next move? (1-9)
2

What is your next move? (1-9)
3

O	O	X
---	---	---

X	X	O
---	---	---

O	X	X
---	---	---

The game is a tie!

Do you want to play again? (yes or no)
no

... Program finished with exit code 0
Press ENTER to exit console.